Wildlife Information System (WIS): Analysis and Design

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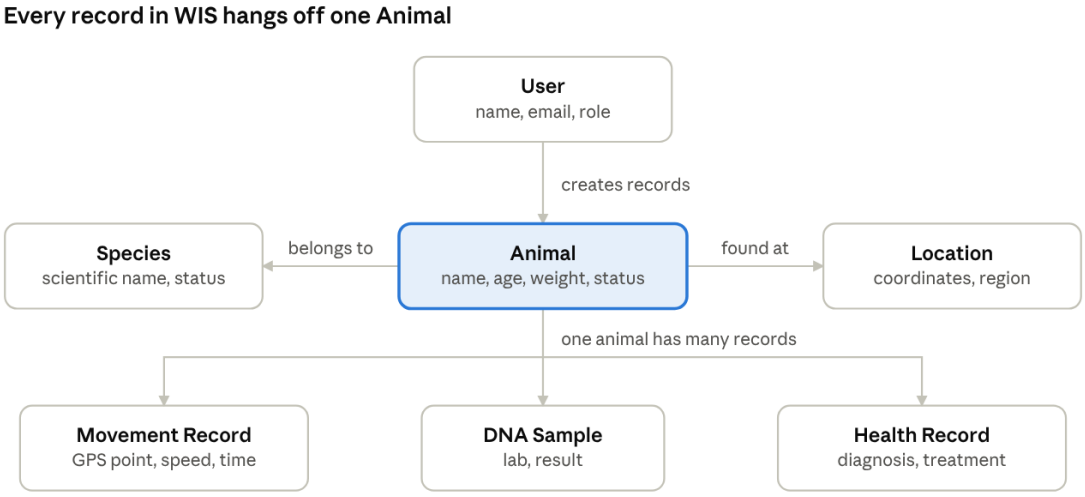
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3. Analysis

This chapter analyses the problem described in Chapter 1: wildlife data is scattered across universities, forest departments and NGOs in paper records, spreadsheets and private databases. WIS stores this data in one central PostgreSQL database and exposes it through a secure RESTful API (JSON) and a basic web interface. The four user roles are Data Entry Operator, Researcher, Government Official and System Administrator. Data arrives from field researchers, IoT tracking devices and lab results. The items Chapter 1 puts out of scope (live tracking maps, automatic alerts, AI/ML, mobile apps, multiple languages) are not analysed here.

3.1 Domain Model

The domain model shows the real-world concepts WIS must record and how they relate, independent of any technology. The Animal is the central concept: it belongs to one Species, is found at a Location, and accumulates Movement Records, DNA Samples and Health Records over time. Users create these records.



domain model · 7 concepts

Users create records; each animal links to one species and one location and owns many movement, DNA and health records.

Domain concepts

Concept	Description	Key attributes
Species	A kind of animal, for example Snow Leopard	common name, scientific name, classification, conservation status, description
Animal	An individual animal tracked by the system	name, gender, date of birth / age, weight (kg), unique marking, status (alive/dead)
Location	A place where animals are found or recorded	name, latitude, longitude, altitude, region, country, habitat type
Movement Record	One GPS reading for an animal	latitude, longitude, altitude, speed (km/h), recorded time, source (for example GPS collar)
DNA Sample	A genetic sample analysed by a lab	sample ID, collection date, sample type, lab name, lab result, species confirmed
Health Record	A veterinary event for an animal	record date, record type, diagnosis, treatment, veterinarian, next follow-up date
User	A person who logs in to WIS	full name, email, role, organization, status

Relationships

From	Relationship	To	Multiplicity
Animal	belongs to	Species	many animals to one species
Animal	is found at	Location	many animals to one location
Animal	has	Movement Record	one animal to zero or more records
Animal	has	DNA Sample	one animal to zero or more samples
Animal	has	Health Record	one animal to zero or more records
User	creates	Movement, DNA and Health records, and animals	one user to many records

Proposed access by role (to be confirmed with stakeholders)

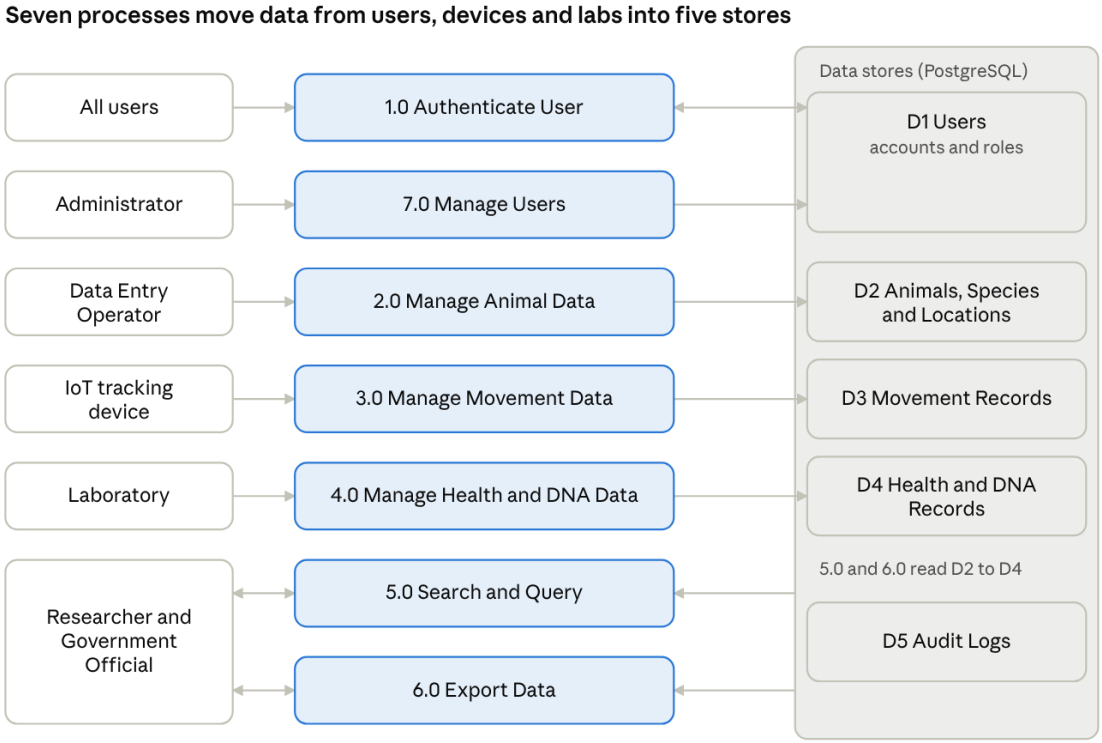
Role	Access
Data Entry Operator	Create and update animal, movement, health and DNA records
Researcher	Search, view and export data through the web interface and API
Government Official	Search, view and export data; view dashboard statistics
System Administrator	Full access, plus user management

3.2 Data Flow Diagram (DFD)

Level 0 (context). WIS is a single process that exchanges data with six external entities.

External entity	Sends to WIS	Receives from WIS
Data Entry Operator	Animal, species, location, health and DNA data; CSV files	Confirmation or validation errors
Researcher	Search filters, API requests, export requests	Result lists, JSON and CSV data
Government Official	Search filters, report requests	Result lists, dashboard statistics, exports
System Administrator	User account details	User list, confirmations
IoT tracking device / field system	GPS readings (source: GPS collar)	Acknowledgement
Laboratory	DNA sample results	Acknowledgement

Level 1. The system is decomposed into seven processes and five data stores.



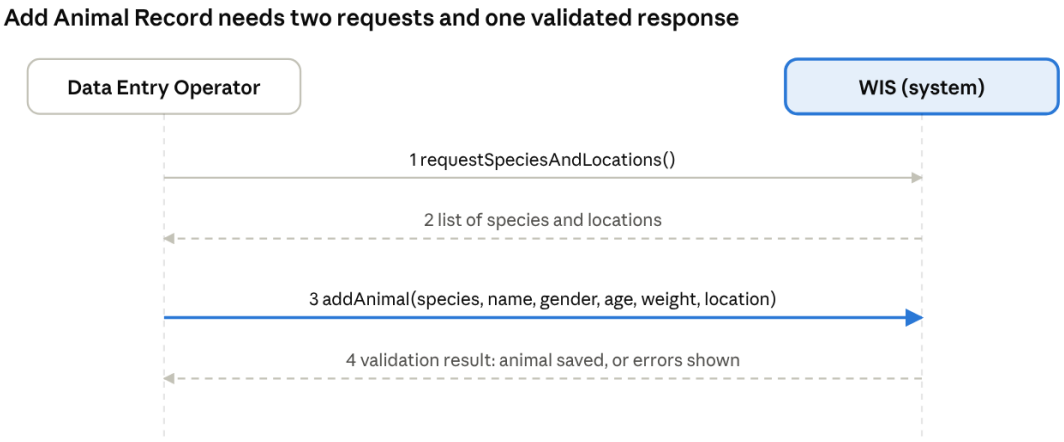
Level 1 DFD · 7 processes, 6 entities, 5 stores

Process	Purpose
1.0 Authenticate User	Checks email and password, issues a JWT, verifies the token and role on every request
2.0 Manage Animal Data	Adds, updates, deletes and views animals, species and locations
3.0 Manage Movement Data	Accepts GPS data from devices and bulk CSV uploads, validates and stores it
4.0 Manage Health and DNA Data	Stores veterinary records and lab DNA results
5.0 Search and Query	Handles filtered searches from the web interface and the REST API
6.0 Export Data	Produces JSON and CSV exports of query results
7.0 Manage Users	Creates, edits and removes user accounts (administrator only)

Data store	Contents
D1 Users	Accounts, password hashes, roles
D2 Animals and Species	Animal profiles, species, locations
D3 Movement Records	GPS readings
D4 Health and DNA Records	Veterinary records and lab DNA results
D5 Audit Logs	Record of who changed what and when

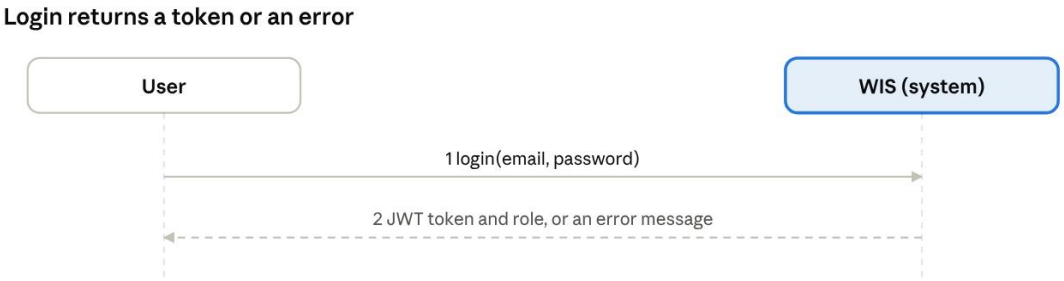
3.3 System Sequence Diagrams

A system sequence diagram treats WIS as a black box and shows only the messages between an actor and the system. The diagram below covers Add Animal Record; the tables cover the other key use cases.



SSD · Add Animal Record · 4 messages

SSD 1: Login



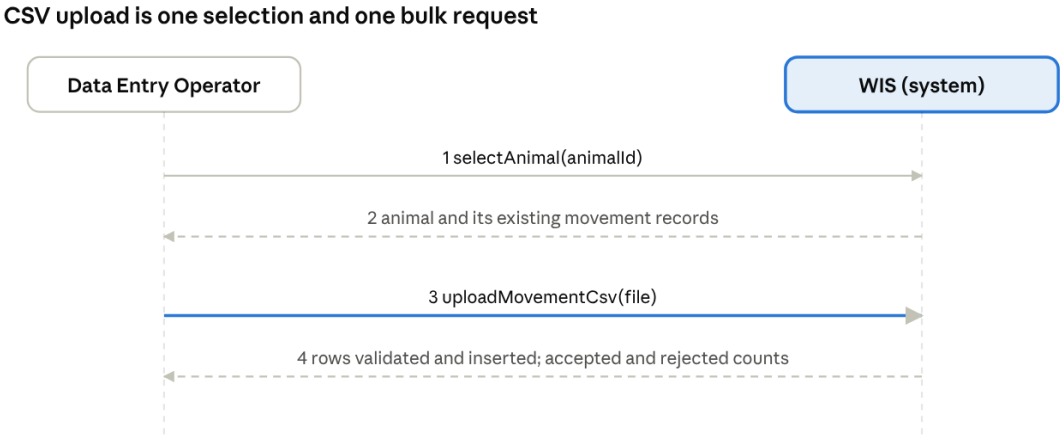
SSD 1 · Login

#	Actor to System	System response
1	login(email, password)	Valid: JWT token and role returned. Invalid: error message

SSD 2: Add Animal Record (diagram above)

#	Actor to System	System response
1	requestSpeciesAndLocations()	List of species and locations
2	addAnimal(species, name, gender, age, weight, location)	Validation result
3	(if valid)	Animal saved, success message with animal ID

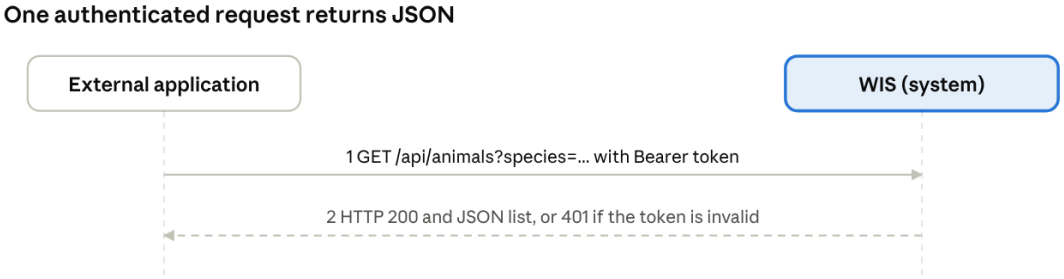
SSD 3: Upload Movement CSV



SSD 3 · Upload Movement CSV

#	Actor to System	System response
1	selectAnimal(animalId)	Animal and existing movement records
2	uploadMovementCsv(file)	Rows validated and inserted; count of accepted and rejected rows

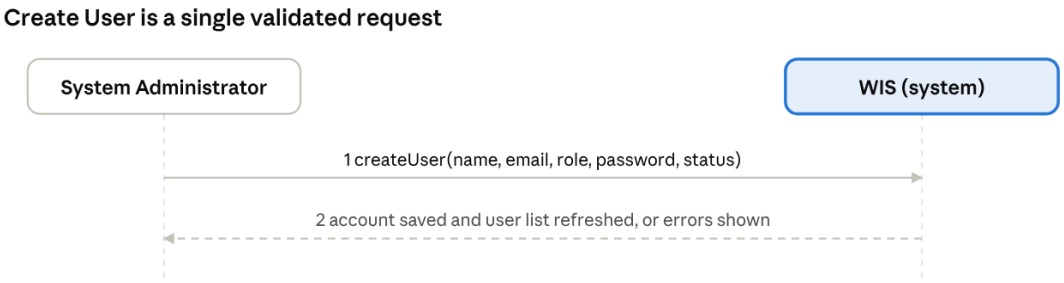
SSD 4: Query Animal Data via API



SSD 4 · Query Animal Data via API

#	Actor to System	System response
1	GET /api/animals?species=... with Bearer token	HTTP 200 and JSON list, or 401 if the token is invalid

SSD 5: Create User (administrator)

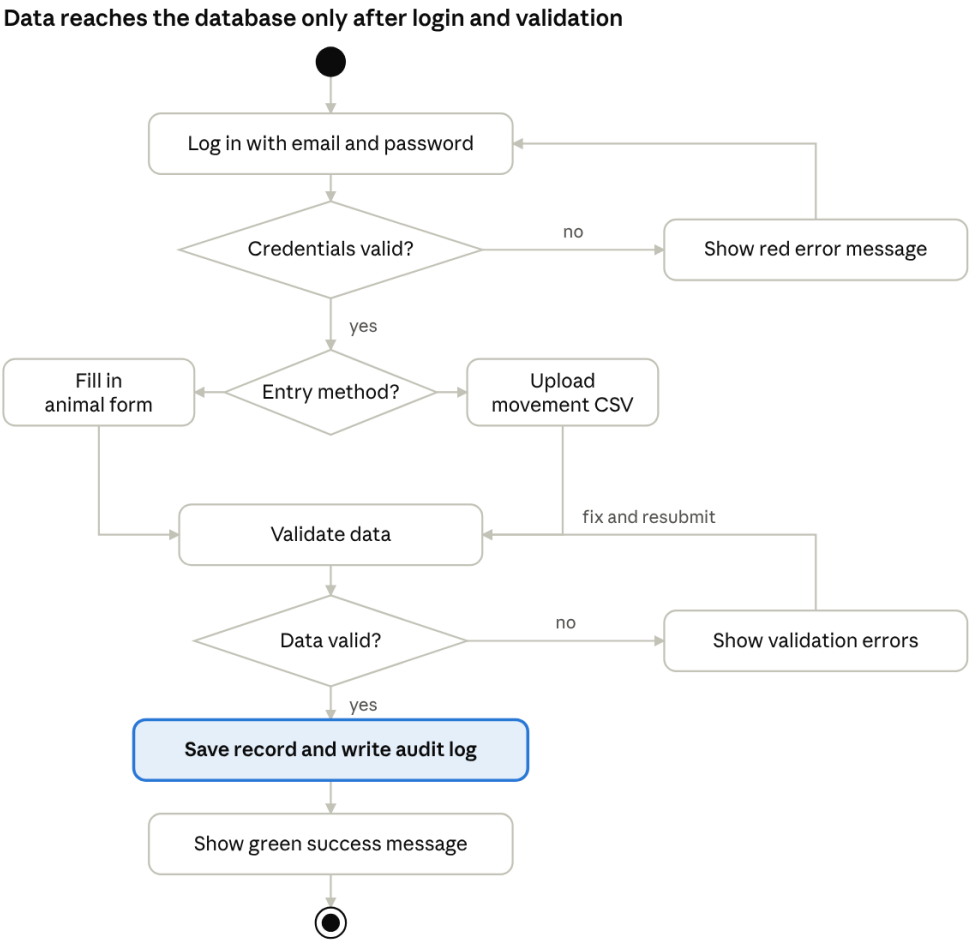


SSD 5 · Create User

#	Actor to System	System response
1	createUser(name, email, role, password, status)	Validation result
2	(if valid)	Account saved, user list refreshed

3.4 Activity Diagram

The activity diagram models the main data-entry workflow: a user logs in, chooses to enter a record manually or upload a CSV file, the system validates the data, and valid data is stored.



activity diagram · 2 decisions, 1 loop each

3.5 Database ER Diagram

The ER diagram maps the domain model to PostgreSQL tables. Primary keys are shown as PK and foreign keys as FK. Each data table also carries a created_by foreign key to users, which supports the audit trail.

↗ ER diagram · 8 tables

Table	Columns
species	species_id (PK), common_name, scientific_name, classification, conservation_status, description
locations	location_id (PK), name, latitude, longitude, altitude, region, country, habitat_type, description
animals	animal_id (PK), species_id (FK), location_id (FK), name, date_of_birth, gender, weight_kg, unique_marking, status, created_by (FK), created_at, updated_at

Table	Columns
movement_records	movement_id (PK), animal_id (FK), location_id (FK), recorded_at, latitude, longitude, altitude, speed_kmh, source, created_at
dna_samples	dna_sample_id (PK), animal_id (FK), sample_id, collection_date, sample_type, lab_name, lab_result, species_confirmed, notes
health_records	health_record_id (PK), animal_id (FK), record_date, record_type, description, veterinarian_name, diagnosis, treatment, next_followup_date, created_at
users	user_id (PK), full_name, email (unique), password_hash, role, organization, status, created_at
audit_logs	log_id (PK), user_id (FK), action, entity, entity_id, logged_at

and success messages in green.